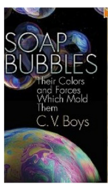
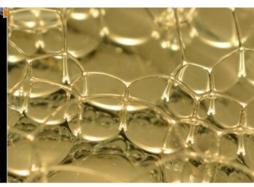
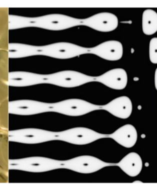
CHE 457: Colloidal & Interfacial Phenomena

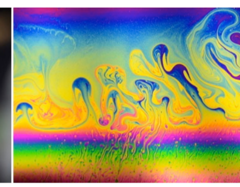
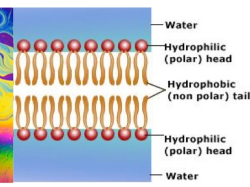
Instructor: Vivek Sharma, Chemical Engineering.

Spring 2026; Monday, Wednesday (& some Fridays), Lecture Hours: 10:00-11:15 AM

Colloidal and Interfacial Phenomena play a central role in many biological, industrial, and environmental processes with applications in food, coatings, pharmaceuticals, cosmetics, lab-on-a-chip, biomedical, petrochemical, energy, and environmental science and technologies. The theory, methods, and applications of *fizzics* (*the scientific study of drops, bubbles, fluid interfaces, emulsions, and foams*) underlie detergency, emulsification, wetting, printing, coating, and brewing. Further application areas include the design of biocompatible materials; lung surfactants; thin films; print electronics and photovoltaics; microfabrication of electronic and electro-optic devices, and enhanced oil recovery. We will delve into fundamental concepts, theoretical methods, and experimental techniques relevant to this interdisciplinary and exciting field, incorporating knowledge and examples from various scientific disciplines, including engineering, physics, chemistry, biology, and medicine. The focus areas in 2026 are: lipids and proteins in biology and plant-based foods; the physics of emulsions; and interfacial flow instabilities.



Course outline & description: The course will cover

- **Colloids and Fizzics in daily life, industry, and biology:** Soap bubbles, soda pop, emulsions, foams, fog, dew, beer froth, shaving foam, whipped cream, molecular gastronomy foams and emulsions, ice cream, lung surfactants, lipids & detergents. What are colloids? “Ben Franklin stilled the waves” & historical perspective. Soap fizzics: detergency, micelles, & bubbles. Dimensional Analysis. (Week 1).
- **Capillarity & deformable interfaces**: Surface tension, contact angle, minimum surfaces. Curvature effects on the equilibrium state of fluids. Young-Laplace equation. Experimental techniques for measuring surface tension. Particles at interfaces. Elasto-capillarity. (Week 2-3)
- **Thermodynamics of interfaces**: Gibbs adsorption equation. Charged interfaces. Debye-Hückel approximation. Colloidal stability. DLVO theory. Brownian motion. Disjoining pressure. Intramolecular & surface forces. Micellization. Solubilization. Proteins: folded, unfolded. (Week 4-6)
- **Physicochemical hydrodynamics**: Wetting & spreading. Fundamentals of nucleation. Capillary condensation. Theories of precipitation. Foam drainage. Thin film hydrodynamics: lipid & soap film drainage. Food emulsions. Emulsions and foams in cosmetics. Interfacial phenomena in printing & coating processes. Spreading, coalescence, and pinching. (Week 7-8)
- **Biological Interfaces**: Lipids & vesicles, proteins at interfaces, cardiovascular bubbles. Analogies between lipid bilayers & soap films. Insoluble monolayers, Langmuir-Blodgett films, & self-assembling monolayers. Liquid crystalline phases. Lung surfactants. (Week 9-11)
- **Special topics:** Food science (plant-based proteins, food hydrocolloids, milk, mayo, cheese, ice cream) OR Biointerfaces (bilayer membranes, cell mechanics, blood, lunch surfactants) OR Interfacial and nonlinear flows. Marangoni effect. Ink-jet printing. Entrainment and drainage. Coating flows. Interfacial rheology. Boussinesq-Scriven model. Instabilities (including Rayleigh-Plateau, Saffman-Taylor, Rayleigh-Bénard, etc.). Viscoelastic interfaces. (Week 12-13)
- **Class projects, discussions & demonstrations**. (Week 14 & Finals).

For graduate students & undergraduates (Check pre-requisites and email queries to: viveks@uic.edu**)**

Grading (most likely): 40% Exams, 20% HWs, 40% Project (Presentation + Report, etc.).